# Functorial Geometry

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# Introduction

Schemes are the central and fundamental objects in algebraic geometry. Classically, they are defined in terms of *locally ringed spaces*, consisting of a topological space and a sheaf of rings whose stalks are local rings. However, the aim of this document is to set up the theory of schemes in an alternative way: the functor-of-points approach. The advantage of this approach is that schemes are defined in terms of (arguably) less data, and that the setup is subjectively cleaner, more categorical and algebraic.

Algebra is fundamentally about solving equations. Given a system of equations, such as

$$\begin{cases} y^2 = x^3 + z \\ z = x + y + 1 \end{cases}$$

one can ask what the set of solutions is. But, what kind of solutions are we talking about? There is a set $X(\mathbb{R})$ of real solutions, a set $X(\mathbb{C})$ of complex solutions, a set $X(\mathbb{F}_q)$ of solutions in every finite field $\mathbb{F}_q$, and more generally, a set $X(R)$ of solutions for every commutative ring $R$. Moreover, note that any morphism $R \to S$ of commutative rings induces a map $X(R) \to X(S)$ between the sets of solutions. Together, the sets of solutions for each commutative ring $R$ form a functor

$$X \colon \mathbf{CRing} \to \mathbf{Set}.$$

Roughly speaking, a *scheme* is such a functor satisfying some properties.

A canonical way to construct such a functor is to consider the functor $\mathrm{Hom}(A, -)$ for a commutative ring $A$. Indeed, any ring morphism $R \to S$ induces a map $\mathrm{Hom}(A, R) \to \mathrm{Hom}(A, S)$. Functors of this form will satsify the properties to be a scheme, and are called *affine* schemes. Let us consider a concrete example.

**Example.** Let $A = \mathbb{Z}[x, y]$ and $X = \mathrm{Hom}(A, -)$. The elements of $X(R)$, which are morphisms $\varphi \colon A \to R$, are determined by a choice of elements $\varphi(x)$ and $\varphi(y)$ in $R$. Hence, $X(R) \cong R^2$ for all commutative rings $R$. In this sense, we can think of $X$ (or of $A$) as a geometric kind of object: the *(affine) plane*. Now suppose that $f \in \mathbb{Z}[x, y]$ is a polynomial and let $A = \mathbb{Z}[x, y]/(f)$ and $X = \mathrm{Hom}(A, -)$. In this case, the elements $\varphi \colon A \to R$ of $X(R)$ are given by pairs $(r, s) = (\varphi(x), \varphi(y)) \in R^2$ such that $f(r, s) = 0$. We can interpret $X$ (or $A$) as a geometric object: the shape described by the equation $f = 0$.

Roughly speaking, the conditions that the functor $X$ needs to satisfy to be a scheme are that it is a

sheaf, and that it 'locally looks like an affine scheme'. In order to make sense of this, we must equip the category **CRing** with a topology, which will be the *Zariski topology*.

It is possible to put different topologies on the category **CRing**, such as the *étale topology*. This will yield the definition of an algebraic space.

# Chapter 1

# Schemes

## 1.1 The Zariski topology

Denote by **CRing** the category of commutative rings. The category of functors $\mathbf{CRing} \to \mathbf{Set}$, that is, the category of presheaves on $\mathbf{CRing}^{\mathrm{op}}$, is denoted by $\mathbf{Set}^{\mathbf{CRing}}$. We will also denote the category $\mathbf{CRing}^{\mathrm{op}}$ by **Aff**.

**Definition 1.1.1.** For any commutative ring $A$, denote by $\operatorname{Spec} A \colon \mathbf{CRing} \to \mathbf{Set}$ the presheaf $\operatorname{Hom}_{\mathbf{CRing}}(A, -)$. This construction is functorial in $A$, and denote by $\operatorname{Spec} \colon \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Set}^{\mathbf{CRing}}$ the resulting functor.

**Definition 1.1.2.** A morphism $f \colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is a *Zariski cover* if for any commutative ring $R$ and element $y \in Y(R)$, there exist elements $r_1, \ldots, r_n \in R$ and elements $x_i \in X(R[r_i^{-1}])$ such that $(r_1, \ldots, r_n) = R$ and $f(R[r_i^{-1}])(x_i) = Y(\gamma_i)(y)$, where $\gamma_i \colon R \to R[r_i^{-1}]$.

**Proposition 1.1.3.** *For any Zariski cover $f \colon X \to Y$ and field $k$, the induced map $X(k) \to Y(k)$ is surjective.*

*Proof.* This follows from the definition as the localization of $k$ (by a non-zero element) is $k$. □

**Proposition 1.1.4.** *(i) For any $X$, the identity morphism $\mathrm{id}_X \colon X \to X$ is a Zariski cover.*

*(ii) The class of Zariski covers is closed under composition: for all Zariski covers $f \colon X \to Y$ and $g \colon Y \to Z$, also $g \circ f$ is a Zariski cover.*

*(iii) If $g \circ f$ is a Zariski cover, then so is $g$.*

*(iv) The class of Zariski covers is closed under pullbacks: for any Zariski cover $f \colon X \to Y$ and any morphism $g \colon Y' \to Y$, also $X \times_Y Y' \to Y'$ is a Zariski cover.*

*(v) If $f \colon X \to Y$ is a Zariski cover and $g \colon Z \to Y$ is a morphism such that $Z \times_Y \operatorname{Spec} R \to \operatorname{Spec} R$ is a Zariski cover for all commutative rings $R$ and $x \in X(R)$, then $g$ is also a Zariski cover.*

*Proof.* *(i)* is obvious (take $n = 1$ and $r_1 = 1$). *(ii)* For any commutative ring $R$ and $z \in Z(R)$, there exist $r_1, \dots, r_n \in R$ and $y_i \in Y(R[r_i^{-1}])$ satisfying the right conditions. Then, for each $i = 1, \dots, n$, there exist $r_{i,1}, \dots, r_{i,m_i} \in R[r_i^{-1}]$ and $x_{i,j} \in X(R[r_i^{-1}][r_{i,j}^{-1}])$ satisfying the right conditions. Note that $R[r_i^{-1}][r_{i,j}^{-1}] \cong R[s_{i,j}]$ for $s_{i,j} = r_{i,j} r_i^k$ some suitably large $k \geq 0$. Hence, the elements $s_{i,j} \in R$ and $x_{i,j} \in X(R[s_{i,j}^{-1}])$ satisfy the right conditions. *(iii)* For any commutative ring $R$ and $z \in Z(R)$, there exist $r_1, \dots, r_n \in R$ and $x_i \in X(R[r_i^{-1}])$ satisfying the right conditions. Then $r_1, \dots, r_n$ and $y_i = f(x_i) \in Y(R[r_i^{-1}])$ also satisfy the right conditions. *(iv)* For any commutative ring $R$ and $y' \in Y'(R)$, let $y = g(y')$. Then there exist $r_1, \dots, r_n \in R$ and $x_i \in X(R[r_i^{-1}])$ satisfying the right conditions. These define elements $x_i' = (x_i, y') \in (X \times_Y Y')(R[r_i^{-1}])$ satisfying the right conditions. *(v)* (Prove this) □

**Definition 1.1.5.** For each commutative ring $R$, denote by $\mathrm{Cov}(R)$ the collection of subfunctors $U \subseteq \operatorname{Spec} R$ such that the inclusion is a Zariski cover. By the above proposition, in particular by *(i)*, *(iv)* and *(v)*, this defines a Grothendieck topology on $\mathbf{CRing}^{\mathrm{op}}$, called the *Zariski topology.*

**Lemma 1.1.6.** *Let $A$ be a commutative ring, and $U \subseteq \operatorname{Spec} A$ a sieve on $A$. Then the following are equivalent:*

*(i) The inclusion $U \hookrightarrow \operatorname{Spec} A$ is a Zariski cover.*

*(ii) There exist elements $a_1, \dots, a_n \in A$ such that $(a_1, \dots, a_n) = A$ and $\gamma_i \in U(A[a_i^{-1}])$, where $\gamma_i \colon A \to A[a_i^{-1}]$ is the localization morphism.*

*Proof.* *(i) ⇒ (ii)* Apply the definition of a Zariski cover to the element $\mathrm{id}_A \in (\operatorname{Spec} A)(A)$.

*(ii) ⇒ (i)* For any commutative ring $R$ and element $f \in (\operatorname{Spec} A)(R)$, there is a commutative diagram

$$\begin{tikzcd} A \arrow[r, "\gamma_i"] \arrow[d, "f"'] & A[a_i^{-1}] \arrow[d] \\ R \arrow[r, "\gamma_i'"] & R[f(a_i)^{-1}] \end{tikzcd}$$

and moreover, $(f(a_1), \dots, f(a_n)) = R$. □

**Proposition 1.1.7.** *The Zariski topology is subcanonical. That is, for any commutative ring $A$, the functor $\operatorname{Spec} A \colon \mathbf{CRing} \to \mathbf{Set}$ is a sheaf on $\mathbf{CRing}^{\mathrm{op}}$.*

*Proof.* Let $a_1, \dots, a_n \in A$ be elements such that $(a_1, \dots, a_n) = A$. We will show that $\operatorname{Spec} A$ satisfies the sheaf condition with respect to the sieve on $A$ generated by $\{A \to A[a_1^{-1}], \dots, A \to A[a_n^{-1}]\}$, that is, we will show that

$$A \to \prod_{i=1}^{n} A[a_i^{-1}] \rightrightarrows \prod_{i,j=1}^{n} A[(a_i a_j)^{-1}] \tag{$*$}$$

is an equalizer diagram. Since the Zariski topology is generated by such sieves (this follows from Lemma 1.1.6), it follows that $\operatorname{Spec} A$ is a sheaf.

To show $A \to \prod_{i=1}^{n} A[a_i^{-1}]$ is injective, suppose the image of $c \in A$ is zero. Then there exist integers $k_i \geq 0$ such that $c a_i^{k_i} = 0$. Since the elements $a_1, \dots, a_n$ generate $A$, there exist elements

$b_i \in A$ such that $a_1 b_1 + \cdots + a_n b_n = 1$. Raising this equality to the power $k_1 + \cdots + k_n$, it follows that $a_1^{k_1} b_1' + \cdots + a_n^{k_n} b_n' = 1$ for some elements $b_i' \in A$. Multiplying by $c$, we find that $c = c a_1^{k_1} b_1' + \cdots + c a_n^{k_n} b_n' = 0$, so the first map in $(*)$ is indeed injective.

To show that $(*)$ is exact in the middle, take any $c = (c_1 a_1^{-k_1}, \ldots, c_n a_n^{-k_n}) \in \prod_{i=1}^n A[a_i^{-1}]$ such that $c_i a_i^{-k_i} = c_j a_j^{-k_j}$ in $A[(a_i a_j)^{-1}]$ for all $i, j$. That is, there exist integers $m_{ij} \geq 0$ such that $(a_i a_j)^{m_{ij}} a_j^{k_j} c_i = (a_i a_j)^{m_{ij}} a_i^{k_i} c_j$ in $A$. Without loss of generality, we may assume $m_{ij} = \max m_{ij} =: m$. As before, there exist $b_1', \ldots, b_n' \in A$ such that $a_1^{m+k_1} b_1' + \cdots + a_n^{m+k_n} b_n' = 1$. Now,

$$a_i^{m+k_i}(a_1^m b_1' c_1 + \cdots + a_n^m b_n' c_n) = a_i^m (a_1^{m+k_1} b_1' + \cdots a_n^{m+k_n} b_n') c_i = a_i^m c_i$$

for all $i$. Hence, $c$ is the image of $a_1^m b_1' c_1 + \cdots + a_n^m b_n' c_n$. Therefore, $(*)$ is an equalizer diagram. $\square$

**Definition 1.1.8.** For any presheaf $X\colon \mathbf{CRing} \to \mathbf{Set}$, denote by $\mathbf{Aff}_X$ the category whose objects are pairs $(R, x)$ consisting of a commutative ring $R$ and an element $x \in X(R)$, and whose morphisms $(R', x') \to (R, x)$ are morphisms $f\colon R \to R'$ such that $x' = X(f)(x)$. Note that an object $(R, x)$ corresponds to morphism $x\colon \operatorname{Spec} R \to X$. In particular, the category $\mathbf{Aff}_X$ is (the opposite of) *the category of elements* of $X$.

(Show that the Zariski topology on $\mathbf{CRing}^{\mathrm{op}}$ naturally induces a Grothendieck topology on $\mathbf{Aff}_X$.)

## 1.2 Affine morphisms and closed immersions

In order to define a scheme, we need the notion of an open immersion in $\mathbf{Set}^{\mathbf{CRing}}$. The idea is to first define closed immersions, and then define an open immersion as the complement of a closed immersion. A related, more general, property of morphisms is that of being affine.

**Definition 1.2.1.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is a *closed immersion* if for every commutative ring $R$ and morphism $\operatorname{Spec} R \to Y$, there is a pullback diagram

$$\begin{tikzcd} \operatorname{Spec} S \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

where $\varphi\colon R \to S$ is a surjective ring morphism.

**Example 1.2.2.** For any ring morphism $\varphi\colon A \to B$, the morphism $\operatorname{Spec}\varphi\colon \operatorname{Spec} B \to \operatorname{Spec} A$ is a closed immersion if and only if $\varphi$ is surjective. Indeed, the 'only if' part follows from taking $R = A$ in the definition. The 'if' part follows from, for any $\operatorname{Spec} R \to \operatorname{Spec} A$, the pullback diagram

$$\begin{tikzcd} \operatorname{Spec} R \otimes_A B \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\ \operatorname{Spec} R \arrow[r] & \operatorname{Spec} A \end{tikzcd}$$

and the fact that the tensor product preserves surjections.

**Definition 1.2.3.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is *affine* if for every commutative ring $R$ and morphism $\operatorname{Spec} R \to Y$, there is a pullback diagram

$$\begin{tikzcd} \operatorname{Spec} S \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

for some commutative ring $S$.

**Proposition 1.2.4.** *(i) Closed immersions are affine morphisms.*

*(ii) Affine morphisms are closed under pullback and composition.*

*(iii) Closed immersions are closed under pullback and composition.*

*(iv) Closed immersions are monomorphisms.*

*Proof.* *(i)*, *(ii)* and *(iii)* are straightforward. For *(iv)*, we must show that, given a closed immersion $f\colon X \to Y$, the induced maps $X(R) \to Y(R)$ are injective for all commutative rings $R$. Suppose that $x, x' \in X(R)$ both map to $y \in Y(R)$. As $f$ is a closed immersion, there is a pullback diagram

$$\begin{tikzcd} \operatorname{Spec} S \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r, "y"] & Y \end{tikzcd}$$

for some surjective morphism $\varphi\colon R \to S$. The two lifts $x, x'\colon \operatorname{Spec} R \to X$ correspond to two sections of $\operatorname{Spec}\varphi$. However, as $\varphi$ is surjective, these sections must agree. $\square$

**Lemma 1.2.5.** *Let $f\colon X \to Y$ be a morphism in $\mathbf{Set}^{\mathbf{CRing}}$ with the property that, for every pullback diagram*

$$\begin{tikzcd} V \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

*the presheaf $V$ is a sheaf. Then, if $Y$ is a sheaf, so is $X$.*

*Proof.* Let $R$ be a commutative ring and $U \subseteq \operatorname{Spec} R$ a covering sieve on $R$. Showing that $X$ is a sheaf amounts to showing that any morphism $\varphi\colon U \to X$ extends uniquely to a morphism $\operatorname{Spec} R \to X$. Since $Y$ is a sheaf, there is a unique morphism $\psi\colon \operatorname{Spec} R \to Y$ such that

$$\begin{tikzcd} U \arrow[r, "\varphi"] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r, "\psi"] & Y \end{tikzcd}$$

commutes. This induces a commutative diagram

$$\begin{tikzcd} U \arrow[r, "\theta"] \arrow[d] & V \arrow[r, "\pi"] \arrow[d, "\rho"] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r, equal] & \operatorname{Spec} R \arrow[r, "\psi"] & Y \end{tikzcd}$$

where $V = X \times_Y \operatorname{Spec} R$. By assumption, $V$ is a sheaf. Hence, there exists a unique morphism $\sigma\colon \operatorname{Spec} R \to V$ extending $\theta$. Now, $\pi \circ \sigma$ is extends $\varphi$. If there is another $\varphi'\colon \operatorname{Spec} R \to X$ extending $\varphi$, then we must have $f \circ \varphi' = \psi$ as $Y$ is a sheaf. But then there exists a unique morphism $\tau\colon \operatorname{Spec} R \to V$ such that $\pi \circ \tau = \varphi'$ and $\rho \circ \tau = \mathrm{id}_R$. Note that, by the universal property of the fiber product, the restriction of $\tau$ to $U$ must be equal to $\theta$. Hence, as $V$ is a sheaf, it follows that $\tau = \sigma$, and hence $\varphi' = \pi \circ \sigma$. Therefore, $X$ is a sheaf. □

**Proposition 1.2.6.** *Let $f\colon X \to Y$ be an affine morphism in $\mathbf{Set}^{\mathbf{CRing}}$.*

*(i) If $Y$ is representable, then so is $X$.*

*(ii) If $Y$ is a sheaf, then so is $X$.*

*Proof.* *(i)* is immediate from the definition. *(ii)* follows from Lemma 1.2.5 and Proposition 1.1.7. □

## 1.3 Open immersions

In some sense, open immersions will be the complement of closed immersions. Locally, it will consist of the points that 'do not intersect' the closed immersion.

**Definition 1.3.1.** For any surjective ring morphism $\varphi\colon A \to B$, denote by $D(\varphi) \subseteq \operatorname{Spec} A$ the subfunctor given by $D(\varphi)(R) = \{f\colon A \to R \mid B \otimes_A R = 0\}$.

**Example 1.3.2.** Let $A$ be a commutative ring and consider a quotient $\varphi\colon A \to A/(a)$ for some element $a \in A$. Note that any ring morphism $f\colon A \to R$ such that $A/(a) \otimes_A R = 0$ lifts uniquely to a morphism $f'\colon A[a^{-1}] \to R$. Conversely, for any ring morphism $f'\colon A[a^{-1}] \to R$, taking the composite with $A \to R$ yields $A/(a) \otimes_A R = 0$. Hence, $D(\varphi) \cong \operatorname{Spec} A[a^{-1}]$ over $\operatorname{Spec} A$.

**Definition 1.3.3.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is an *open immersion* if for every commutative ring $R$ and morphism $\operatorname{Spec} R \to Y$, there is a pullback diagram

$$\begin{tikzcd} D(\varphi) \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

where $\varphi\colon R \to S$ is some surjective ring morphism.

**Lemma 1.3.4.** *Let $\varphi\colon A \to B$ be a surjective ring morphism. Then the inclusion $D(\varphi) \hookrightarrow \operatorname{Spec} A$ is an open immersion.*

*Proof.* Consider any pullback diagram

$$\begin{tikzcd} U \arrow[r] \arrow[d, hook] & D(\varphi) \arrow[d, hook] \\ \operatorname{Spec} R \arrow[r, "\operatorname{Spec} g"] & \operatorname{Spec} A \end{tikzcd}$$

for some ring morphism $g\colon A \to R$. Then $U \subseteq \operatorname{Spec} R$ is the subfunctor given by

$$U(S) = \{f\colon R \to S \mid B \otimes_A S = 0\} = \{f\colon R \to S \mid (B \otimes_A R) \otimes_R S = 0\}.$$

Therefore, $U = D(\psi)$ where $\psi\colon R \to B \otimes_A R$ is the base change of $\varphi$. □

**Proposition 1.3.5.** *Let $\varphi\colon A \to B$ be a surjective ring morphism. Then $D(\varphi)$ is a sheaf.*

*Proof.* Let $R$ be a commutative ring and $U \subseteq \operatorname{Spec} R$ a covering sieve on $R$. Showing that $D(\varphi)$ is a sheaf amounts to showing that any morphism $s\colon U \to \operatorname{Spec} R$ extends uniquely to a morphism $\operatorname{Spec} R \to D(\varphi)$. Since $\operatorname{Spec} A$ is a sheaf, there exists a unique ring morphism $g\colon A \to R$ such that such that the diagram

$$\begin{tikzcd} U \arrow[r, "s"] \arrow[d, hook] & D(\varphi) \arrow[d, hook] \\ \operatorname{Spec} R \arrow[r, "\operatorname{Spec} g"] & \operatorname{Spec} A \end{tikzcd}$$

commutes. Our goal is now to show that $g \in D(\varphi)(R)$. As in the proof of Lemma 1.3.4, we have a pullback diagram

$$\begin{tikzcd} D(\psi) \arrow[r] \arrow[d, hook] & D(\varphi) \arrow[d, hook] \\ \operatorname{Spec} R \arrow[r, "\operatorname{Spec} g"] & \operatorname{Spec} A \end{tikzcd}$$

where $\psi\colon R \to S := B \otimes_A R$ is the base change of $\varphi$. Note that $U \subseteq D(\psi) \subseteq \operatorname{Spec} R$. Let $V \subseteq \operatorname{Spec} S$ be the pullback of $U \subseteq \operatorname{Spec} R$ along $\operatorname{Spec} \psi$. Then $V$ is a covering sieve on $S$ as $U$ is a covering sieve on $R$. Now, for any ring morphism $t\colon S \to T$ such that $t \in V(T)$, we have $t \circ \psi \in D(\psi)$, that is, $(S \otimes_R S) \otimes_S T = 0$, so $T = 0$ (explain why $T = 0$). Therefore, $V$ is either empty or generated by $\{S \to 0\}$, and thus $S = 0$. This shows that $g \in D(\varphi)(R)$, which concludes the proof. □

**Corollary 1.3.6.** *Let $f\colon X \to Y$ be an open immersion in $\mathbf{Set}^{\mathbf{CRing}}$. If $Y$ is a sheaf, then so is $X$.*

*Proof.* Follows directly from Lemma 1.2.5 and Proposition 1.3.5. □

**Example 1.3.7.** $D(\varphi)$ need not be representable. For example, consider any morphism $\varphi\colon \mathbb{Z}[x, y] \to \mathbb{Z}$. (Show that $D(\varphi)$ is not representable)

**Proposition 1.3.8.** *(i) Open immersions are closed under pullback.*

*(ii) Open immersions are monomorphisms.*

*(iii) Given morphisms $f\colon X \to Y$ and $g\colon Y \to Z$ in $\mathbf{Set}^{\mathbf{CRing}}$ such that $g$ is a morphisms $g \circ f$ an open immersion, then $f$ is also an open immersion.*

*Proof.* *(i)* is straightforward. For *(ii)*, we must show that, given an open immersion $f\colon X \to Y$, the induced maps $X(R) \to Y(R)$ are injective for all commutative rings $R$. Suppose that $x, x' \in X(R)$ both map to $y \in Y(R)$. As $f$ is an open immersion, there is a pullback diagram

$$\begin{tikzcd} D(\varphi) \arrow[r] \arrow[d, "i"] & X \arrow[d] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

for some surjective morphism $\varphi\colon R \to S$. The two lifts $x, x'\colon \operatorname{Spec} R \to X$ correspond to two sections of $i$. However, as $i$ is injective, we must have $D(\varphi) = \operatorname{Spec} R$ and these sections must agree. For *(iii)*, as $g$ is a monomorphism, the diagram

$$\begin{tikzcd} X \arrow[r, equal] \arrow[d, "f"'] & X \arrow[d, "g \circ f"] \\ Y \arrow[r, "g"] & Z \end{tikzcd}$$

is a pullback diagram, so the result follows from *(i)*. □

## 1.4 Schemes

We are now ready to define what a scheme is.

**Definition 1.4.1.** A *scheme* is a sheaf $X\colon \mathbf{CRing} \to \mathbf{Set}$ for which there exists a set $\mathcal{U}$ of pairs $(R, x)$ consisting of a commutative ring $R$ and an element $x \in X(R)$ such that

- for all $(R, x) \in \mathcal{U}$, the corresponding morphism $\operatorname{Spec} R \xrightarrow{x} X$ is an open immersion,
- the induced morphism $\coprod_{(R,x)\in\mathcal{U}} \operatorname{Spec} R \to X$ is a Zariski cover.

Such a set $\mathcal{U}$ is called an *atlas* for $X$. The full subcategory of $\mathbf{Set}^{\mathbf{CRing}}$ of schemes is denoted $\mathbf{Sch}$.

**Example 1.4.2.** For any commutative ring $A$, the sheaf $\operatorname{Spec} A$ is a scheme. Indeed, we can simply take $\mathcal{U} = \{(A, \mathrm{id}_A)\}$ as atlas. In particular, Spec descends to a functor $\mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$.

**Definition 1.4.3.** A scheme $X$ is *affine* if it is representable as a functor, that is, if $X \cong \operatorname{Spec} A$ for some commutative ring $A$.

**Proposition 1.4.4.** *Let $X$ be a scheme. For every field $k$ and element $x \in X(k)$, the morphism $\operatorname{Spec} k \xrightarrow{x} X$ factors through an open immersion $\operatorname{Spec} R \to X$.*

*Proof.* There exists an atlas $\mathcal{U}$ of $X$ such that $\coprod_{(R,x)\in\mathcal{U}} \operatorname{Spec} R \to X$ is a Zariski cover. By Proposition 1.1.3, the morphism $\operatorname{Spec} k \xrightarrow{x} X$ factors through this morphism. □

**Proposition 1.4.5.** *Let $\varphi\colon A \to B$ be a surjective ring morphism. For every $a \in A$, write $\gamma_a\colon A \to A[a^{-1}]$ for the localization morphism. Then $D(\varphi)$ is a scheme with atlas $\mathcal{U} = \{(A[a^{-1}], \gamma_a)\colon a \in \ker\varphi\}$.*

*Proof.* First, $D(\varphi)$ is a sheaf by Proposition 1.3.5. For any $a \in \ker\varphi$, the morphism $\operatorname{Spec} A[a^{-1}] \to D(\varphi)$ given by $\gamma_a$ is an open immersion by Proposition 1.3.8 *(iii)* as $\operatorname{Spec} A[a^{-1}] \to \operatorname{Spec} A$ is an open immersion (see Example 1.3.2). It remains to show that $\coprod_{a\in\ker\varphi} \operatorname{Spec} A[a^{-1}] \to D(\varphi)$ is a Zariski cover. So, take any commutative ring $R$ and element $g \in D(\varphi)(R)$. For any $a \in \ker\varphi$, we have a pushout diagram

$$\begin{tikzcd} A \arrow[r, "\gamma_a"] \arrow[d, "g"'] & A[a^{-1}] \arrow[d, "g[a^{-1}]"] \\ R \arrow[r, "\gamma_{g(a)}"] & R[g(a)^{-1}] \end{tikzcd}$$

so $\gamma_{g(a)} \circ g\colon A \to R[g(a)^{-1}]$ is an element of $D(\varphi)(R[g(a)^{-1}])$. Since $B \otimes_A R = 0$, we have $g(\ker\varphi) = R$, so there exist elements $a_1, \ldots, a_n \in \ker\varphi$ such that $g(a_1), \ldots, g(a_n)$ generate $R$. Now the elements $g(a_1), \ldots, g(a_n) \in R$ together with the elements $g[a_i^{-1}] \in (\coprod_{a \in \ker\varphi} \operatorname{Spec} A[a^{-1}])(R[g(a_i)^{-1}])$ satisfy the definition of a Zariski cover. $\square$

**Proposition 1.4.6.** *Open immersions are closed under composition.*

*Proof.* Let $f\colon X \to Y$ and $g\colon Y \to Z$ be open immersions. For any commutative ring $R$ and morphism $\operatorname{Spec} R \to Z$, we construct the diagram

$$\begin{tikzcd} U \arrow[r] \arrow[d, hook] & X \arrow[d, "f"] \\ V \arrow[r] \arrow[d, hook] & Y \arrow[d, "g"] \\ \operatorname{Spec} R \arrow[r] & Z \end{tikzcd}$$

in which both squares are pullback squares. By Corollary 1.3.6, both $U$ and $V$ are sheaves. Define

$$I = \{a \in A \mid \gamma_a \in U(A[a^{-1}])\}$$
$$J = \{a \in A \mid \gamma_a \in V(A[a^{-1}])\}$$

and note that $I$ and $J$ are ideals with $I \subseteq J$. (Explain why $I$ and $J$ are they ideals, using that $U$ and $V$ are sheaves) Define ring morphisms $\varphi\colon A \to A/I$ and $\psi\colon A \to A/J$, and note that $V = D(\psi)$. Our goal is to show that $U = D(\varphi)$. For all $a \in J$, let $U_a \hookrightarrow \operatorname{Spec} A[a^{-1}]$ be the pullback of $U \hookrightarrow V$ along $\operatorname{Spec} A[a^{-1}] \xrightarrow{\gamma_a} V$. Then we obtain a pullback diagram

$$\begin{tikzcd} \coprod_{a \in J} U_a \arrow[r] \arrow[d, hook] & U \arrow[d, hook] \\ \coprod_{a \in J} \operatorname{Spec} A[a^{-1}] \arrow[r] & V \end{tikzcd}$$

in which the bottom horizontal morphism is a Zariski cover, and therefore the top horizontal morphism is also a Zariski cover. For each $a \in J$, define

$$I_a = \{b \in A[a^{-1}] \mid \gamma_b \in U_a(A[a^{-1}][b^{-1}])\}.$$

Since $U_a \hookrightarrow \operatorname{Spec} A[a^{-1}]$ is an open immersion (being the pullback of an open immersion), it follows that the induced morphism $\coprod_{b \in I_a} \operatorname{Spec} A[a^{-1}][b^{-1}] \to U_a$ is a Zariski cover, and hence the composite

$$\coprod_{a \in J} \coprod_{b \in I_a} \operatorname{Spec} A[a^{-1}][b^{-1}] \to \coprod_{a \in J} U_a \to U$$

is a Zariski cover (see Proposition 1.1.4). Now, for every $a \in J$ and $b \in I_a$, there exists some $c \in A$ such that

$$\begin{tikzcd} A \arrow[r, "\gamma_c"] \arrow[d, "\gamma_a"'] & A[c^{-1}] \arrow[d, equal, "\wr"] \\ A[a^{-1}] \arrow[r, "\gamma_b"] & A[a^{-1}][b^{-1}] \end{tikzcd}$$

commutes, and hence $c \in I$. Therefore, the induced map $\coprod_{c \in I} \operatorname{Spec} A[c^{-1}] \to U$ is a Zariski cover, and we conclude that $U = D(\varphi)$. $\square$

**Lemma 1.4.7.** *Let $f\colon X \to Y$ be a morphism in $\mathbf{Set}^{\mathbf{CRing}}$. Suppose $Y$ is a scheme with atlas $\mathcal{U}$ such that for every $(R, y) \in \mathcal{U}$ there is a pullback diagram*

$$\begin{tikzcd} U_{(R,y)} \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r, "y"] & Y \end{tikzcd}$$

*with $U_{(R,y)}$ a scheme. Then, if $X$ is a sheaf, it is also a scheme.*

*Proof.* For every $(R, y) \in \mathcal{U}$, let $\mathcal{U}_{(R,y)}$ be an atlas for $U_{(R,y)}$. Note that the morphism $U_{(R,y)} \to X$, being the pullback of $y$, is an open immersion, so, in particular, $U_{(R,y)}$ can be identified with a subfunctor of $X$. We will show that $\mathcal{U}_X = \bigcup_{(R,y)\in\mathcal{U}} \mathcal{U}_{(R,y)}$ is an atlas for $X$. Indeed, for every $(S, x) \in \mathcal{U}_X$, the morphism $\operatorname{Spec} S \xrightarrow{x} X$, being the composition of two open immersions, is an open immersion by Proposition 1.4.6. Furthermore, $\coprod_{(S,x)\in\mathcal{U}_X} \operatorname{Spec} S \to X$ is a Zariski cover because $\coprod_{(R,y)\in\mathcal{U}} \operatorname{Spec} U_{(R,y)} \to X$ is a Zariski cover and each $\coprod_{(S,x)\in\mathcal{U}_{(R,y)}} \operatorname{Spec} S \to U_{(R,y)}$ is a Zariski cover (see Proposition 1.1.4). □

**Corollary 1.4.8.** *Let $f\colon X \to Y$ be a morphism in $\mathbf{Set}^{\mathbf{CRing}}$ with $Y$ a scheme.*

*(i) If $f$ is affine, then $X$ is a scheme.*

*(ii) If $f$ is a closed immersion, then $X$ is a scheme.*

*(iii) If $f$ is an open immersion, then $X$ is a scheme.*

*Proof.* All cases are special cases of Lemma 1.4.7. For *(i)*, the $U_{(R,y)}$ are affine and hence schemes. *(ii)* follows from *(i)* as closed immersions are affine. *(iii)* follows from Proposition 1.4.5. □

Another way to construct schemes out of other schemes is via fiber products.

**Proposition 1.4.9.** *The category* **Sch** *is closed under fiber products.*

*Proof.* Let $f\colon X \to Z$ and $g\colon Y \to Z$ be morphisms of schemes. As $X, Y$ and $Z$ are sheaves, so is the fiber product $X \times_Z Y$. Let $\mathcal{U}_Z = \{(C_i, z_i) : i \in I\}$ be an atlas for $Z$. As the $z_i\colon \operatorname{Spec} C_i \to Z$ are open immersions, so are the pullbacks $x_i\colon X \times_Z \operatorname{Spec} C_i \to X$ and $y_i\colon Y \times_Z \operatorname{Spec} C_i \to Y$. By Corollary 1.4.8 *(iii)*, also $X \times_Z \operatorname{Spec} C_i$ and $Y \times_Z \operatorname{Spec} C_i$ are schemes, so they have atlases $\mathcal{U}_{X,i} = \{(A_{ij}, x_{ij}) : j \in J_i\}$ and $\mathcal{U}_{Y,i} = \{(B_{ik}, y_{ik}) : k \in K_i\}$, respectively. Now it is clear that

$$\mathcal{U} = \{(A_{ij} \otimes_{C_i} B_{ik}, (x_{ij}, y_{ik})) : i \in I \text{ and } (j, k) \in J_i \times K_i\}$$

is an atlas for $X \times_Z Y$, so it is also a scheme. □

**Example 1.4.10.** For any $n \geq 0$, the affine scheme $\mathbb{A}^n = \operatorname{Spec} \mathbb{Z}[x_1, \ldots, x_n]$ is called *affine $n$-space*.

**Proposition 1.4.11.** *The functor* $\operatorname{Spec}\colon \mathbf{CRing}^{\mathrm{op}} \to \mathbf{Sch}$ *has a left adjoint* $\Gamma\colon \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$*, called the* global functions functor*. In particular, for any scheme* $X$ *and commutative ring* $A$*, there is a bijection*

$$\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} A) \cong \operatorname{Hom}_{\mathbf{CRing}}(A, \Gamma(X))$$

*natural in* $X$ *and* $A$*.*

*Proof.* Any presheaf $X$ is the colimit of representable presheaves, that is, $X = \operatorname{colim}(\operatorname{Spec} \circ F)$ where $F\colon \mathbf{Aff}_X \to \mathbf{CRing}^{\mathrm{op}}$ is the forgetful functor which sends $(R, x)$ to $R$. For simplicity, we will write this as $X = \operatorname{colim}_{\operatorname{Spec} R \to X} \operatorname{Spec} R$. Now we define $\Gamma(X) = \operatorname{colim} F$, which extends to a functor $\Gamma\colon \mathbf{Sch} \to \mathbf{CRing}^{\mathrm{op}}$. Now, we obtain bijections

$$\begin{aligned}\operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} A) &= \operatorname{Hom}_{\mathbf{Sch}}(\operatorname*{colim}_{\operatorname{Spec} R \to X} \operatorname{Spec} R, \operatorname{Spec} A) = \lim_{\operatorname{Spec} R \to X} \operatorname{Hom}_{\mathbf{Sch}}(\operatorname{Spec} R, \operatorname{Spec} A) \\ &= \lim_{\operatorname{Spec} R \to X} \operatorname{Hom}_{\mathbf{CRing}^{\mathrm{op}}}(R, A) = \operatorname{Hom}_{\mathbf{CRing}^{\mathrm{op}}}(\operatorname*{colim}_{\operatorname{Spec} R \to X} R, A) = \operatorname{Hom}_{\mathbf{CRing}}(A, \Gamma(X))\end{aligned}$$

which are natural in $X$ and $A$. □

**Proposition 1.4.12.** *There is a natural isomorphism between* $\Gamma$ *and* $\operatorname{Hom}_{\mathbf{Sch}}(-, \mathbb{A}^1)$ *as functors* $\mathbf{Sch} \to \mathbf{Set}$*.*

*Proof.* For any scheme $X$, note that elements of $\Gamma(X)$ can be identified with morphisms $\mathbb{Z}[x] \to \Gamma(X)$, as the image of $x$. The adjunction $\Gamma \dashv \operatorname{Spec}$ gives the isomorphism

$$\operatorname{Hom}_{\mathbf{CRing}}(\mathbb{Z}[x], X) \cong \operatorname{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec} \mathbb{Z}[x]) = \operatorname{Hom}_{\mathbf{Sch}}(X, \mathbb{A}^1),$$

which is natural in $X$. □

## 1.5 The topological space of a scheme

**Definition 1.5.1.** Let $X$ be a scheme. Given fields $k$ and $k'$, we say that morphisms $x\colon \operatorname{Spec} k \to X$ and $x'\colon \operatorname{Spec} k' \to X$ are *equivalent* if there exists a field $\ell$ and a commutative diagram

$$\begin{tikzcd} \operatorname{Spec} \ell \arrow[r] \arrow[d] & \operatorname{Spec} k' \arrow[d, "x'"] \\ \operatorname{Spec} k \arrow[r, "x"] & X \end{tikzcd}.$$

One easily shows that this defines an equivalence relation on the set of morphisms from spectra of fields to $X$. The set of equivalence classes is denoted by $|X|$.

Given a morphism of schemes $f\colon X \to Y$, post-composition with $f$ induces a map $|f|\colon |X| \to |Y|$. In particular, we obtain a functor

$$|-|\colon \mathbf{Sch} \to \mathbf{Set}.$$

Now, we want to define a topology on $|X|$. In order to do this, we first consider the affine case. We first define the topology locally, then extend it to globally.

**Proposition 1.5.2.** *If $X$ is affine, that is, $X = \operatorname{Spec} R$, then $|X|$ can be identified with the set of prime ideals of $R$.*

*Proof.* A morphism $x\colon \operatorname{Spec} k \to X$ corresponds to a ring morphism $R \to k$, whose kernel is a prime ideal. Note that any equivalent morphism $x'\colon \operatorname{Spec} k' \to X$ yields the same prime ideal, since field extensions $k \subseteq \ell \supseteq k'$ are injective. Conversely, a prime ideal $\mathfrak{p}$ of $R$ induces a ring morphism $R \to R/\mathfrak{p} \to Q(R/\mathfrak{p})$, where $R/\mathfrak{p}$ is a domain and $Q(R/\mathfrak{p})$ its field of fractions. These two constructions are inverse to each other. □

However, we don't really want prime ideals to be the focus. Instead, we focus on points. We define closed subsets to be sets of points that vanish under ideals.

**Definition 1.5.3.** Let $X = \operatorname{Spec} R$ be an affine scheme. The *Zariski topology* on $|X|$ is the topology whose closed sets are precisely the sets $V(I) = \{x \in |X| \mid x(I) = 0\}$ for some ideal $I \subseteq R$. Indeed, this defines a topology as

- $|\operatorname{Spec} R| = V(0)$ and $\varnothing = V(R)$ are both closed,
- $V(I) \cup V(J) = V(IJ)$ for all ideals $I, J \subseteq R$,
- $\bigcap_\alpha V(I_\alpha) = V(\sum_\alpha I_\alpha)$ for all ideals $I_\alpha \subseteq R$.

**Definition 1.5.4.** Let $X$ be any scheme. The *Zariski topology* on $|X|$ is the finest topology such that $|f|$ is continuous for all morphisms $f\colon \operatorname{Spec} R \to X$. That is, a set $U \subseteq |X|$ is open (resp. closed) if and only if $|f|^{-1}U$ is open (resp. closed) in $|\operatorname{Spec} R|$ for all morphisms $f\colon \operatorname{Spec} R \to X$.

**Remark 1.5.5.** When $X = \operatorname{Spec} R$, the more general definition agrees with the special case. Indeed, for any closed set $V(I) \subseteq |X|$ and any morphism $f\colon \operatorname{Spec} S \to \operatorname{Spec} R$ (dually $\varphi\colon R \to S$), we have $|f|^{-1}V(I) = V(J)$ for the ideal $J = S\varphi(I) \subseteq S$. Conversely, if $V \subseteq |X|$ is such that $|f|^{-1}V$ is closed for all morphisms $f\colon \operatorname{Spec} S \to \operatorname{Spec} R$, then specializing to $f = \mathrm{id}_X$ gives that $V = V(I)$ for some ideal $I \subseteq R$.

**Lemma 1.5.6.** *Any morphism $f\colon X \to Y$ of schemes induces a continuous map $|f|\colon |X| \to |Y|$.*

*Proof.* Let $U \subseteq |Y|$ be an open set. Then $|g|^{-1}(|f|^{-1}U) = |f \circ g|^{-1}U \subseteq |\operatorname{Spec} R|$ is open for any morphism $g\colon \operatorname{Spec} R \to X$. Hence, $|f|^{-1}U \subseteq |X|$ is open. □

In particular, we have promoted the functor to

$$|-|\colon \mathbf{Sch} \to \mathbf{Top}.$$

**Proposition 1.5.7.** *Let $f\colon X \to Y$ be a closed immersion. Then $|f|$ provides a homeomorphism of $|X|$ onto a closed subset of $|Y|$.*

*Proof.* (Write proof) □

**Definition 1.5.8.** Let $X$ be a scheme with a point $x \in |X|$ given by $x\colon \operatorname{Spec} k \to X$. If $\operatorname{Spec} R \to X$ is an affine open containing $x$, then the *local ring* of $X$ at $x$, denoted $\mathcal{O}_{X,x}$, is the localization $S^{-1}R$ where $S = \{f \in R \mid x(f) \neq 0\}$. This is, up to canonical isomorphism, independent of the chosen open affine $\operatorname{Spec} R$.

## 1.6 Properties of schemes

### 1.6.1 Quasi-compact schemes

**Definition 1.6.1.** A scheme $X$ is *quasi-compact* if the topological space $|X|$ is compact.

**Lemma 1.6.2.** *An affine scheme is quasi-compact.*

*Proof.* Let $|X| = \bigcup_i U_i$ be an open cover, with $U_i = |X| \setminus V(I_i)$ for some ideals $I_i \subseteq R$. Note that the ideal $J$ generated by the $I_i$ is equal to $R$. Indeed, if $J \subsetneq R$, then $J$ is contained in some maximal ideal $\mathfrak{m}$ of $R$, but then the point $x \in |X|$ given by $x\colon R \to R/\mathfrak{m}$ would not be contained in any of the $U_i$. Now, the $I_i$ generate $R$, but then so do finitely many $I_i$, and in turn $|X|$ is covered by finitely many of the $U_i$. □

**Example 1.6.3.** (Show that the complement of the origin in $\operatorname{Spec} k[x_1, x_2, \ldots]$ is not quasi-compact)

### 1.6.2 Connected schemes

**Definition 1.6.4.** A scheme $X$ is *connected* if the topological space $|X|$ is connected.

### 1.6.3 Irreducible schemes

**Definition 1.6.5.** A scheme $X$ is *irreducible* if the topological space $|X|$ is irreducible.

### 1.6.4 Reduced schemes

**Definition 1.6.6.** A scheme $X$ is *reduced* if for all commutative rings $R$ and open immersions $\operatorname{Spec} R \to X$, the ring $R$ is reduced.

**Proposition 1.6.7.** *Let $X$ be a scheme. The following are equivalent:*

*(i) $X$ is reduced,*

*(ii) for all $x \in |X|$, the local rings $\mathcal{O}_{X,x}$ are reduced.*

*Proof.* Note that *(i)* $\Rightarrow$ *(ii)* as the localization of a reduced ring is again reduced. For *(ii)* $\Rightarrow$ *(i)*, suppose $\operatorname{Spec} R \to X$ is an open immersion and $f \in R$ is nilpotent, that is, $f^n = 0$ for some $n \geq 1$.

Consider the ideal $I = \{r \in R \mid rf^n = 0\}$ of $R$, which we claim is equal to $R$. Namely, if $I \subsetneq R$, then $I$ is contained in a maximal ideal $\mathfrak{m}$ of $R$, and we consider the local ring $\mathcal{O}_{X,x} = R_{\mathfrak{m}}$ for the corresponding point $x\colon \operatorname{Spec}(R/\mathfrak{m}) \to X$. By assumption, we must have that the image of $f^n$ in $R_{\mathfrak{m}}$ is zero, that is, $sf^n = 0$ for some $s \notin \mathfrak{m}$, but then $s \in I \subseteq \mathfrak{m}$, a contradiction. Hence, $I = R$ and thus $f^n = 0$. □

### 1.6.5 Integral schemes

**Definition 1.6.8.** A scheme $X$ is *integral* if for all commutative rings $R$ and open immersions $\operatorname{Spec} R \to X$, the ring $R$ is a domain.

**Proposition 1.6.9.** *Let $X$ be a scheme. The following are equivalent:*

*(i) $X$ is integral,*

*(ii) $X$ is reduced and irreducible,*

*Proof. (i) ⇒ (ii)* As domains are reduced, being integral implies being reduced. To show $|X|$ is irreducible, we first assume that $X$ is affine, that is, $X = \operatorname{Spec} R$. Suppose $|X| = Y \cup Z$ for some closed subsets $Y = V(I)$ and $Z = V(J)$, that is, $IJ = 0$. Since $R$ is a domain, this implies $I = 0$ or $J = 0$, that is, $Y = |X|$ or $Z = |X|$. For a general scheme $X$, (reduce to the local case). *(ii) ⇒ (i)* Let $\operatorname{Spec} R \to X$ be an open immersion. By assumption $R$ is reduced, and $\operatorname{Spec} R$ is irreducible as an open subset of an irreducible topological space is again irreducible. Now, to show that $R$ is a domain, suppose $fg = 0$ for some non-zero $f, g \in R$. Then $\operatorname{Spec} R = V((f)) \cup V((g))$, so we must have $V((f)) = \operatorname{Spec} R$ or $V((g)) = \operatorname{Spec} R$, implying that $f$ or $g$ is nilpotent. But since $R$ is reduced, this implies $f$ or $g$ is zero, so $R$ is a domain. □

### 1.6.6 Noetherian schemes

**Definition 1.6.10.** A scheme $X$ is *locally noetherian* if for all commutative rings $R$ and open immersions $\operatorname{Spec} R \to X$, the ring $R$ is noetherian. We say $X$ is *noetherian* if it is locally noetherian and quasi-compact.

**Example 1.6.11.** For any noetherian ring $A$, the scheme $\coprod_{i=1}^{\infty} \operatorname{Spec} A$ is locally noetherian, but not noetherian as it is not quasi-compact.

### 1.6.7 Normal schemes

**Definition 1.6.12.** (Define normal schemes)

### 1.6.8 Regular schemes

**Definition 1.6.13.** (Define regular schemes)

## 1.7 Properties of morphisms of schemes

### 1.7.1 Quasi-compact morphisms

**Definition 1.7.1.** A morphism of schemes $f\colon X \to Y$ is *quasi-compact* if for every open immersion $\operatorname{Spec} R \to Y$, the fiber product $\operatorname{Spec} R \times_Y X$ is quasi-compact.

**Proposition 1.7.2.** *The following are equivalent:*

*(i)* $f\colon X \to Y$ *is quasi-compact.*

*(ii) There exists an affine open covering such that* (...)

*(iii) If the induced map* $|f|\colon |X| \to |Y|$ *is proper.*

(Closed under pullback, closed under composition)

**Lemma 1.7.3.** *Being a quasi-compact morphism is closed under pullback and composition.*

*Proof.* Closed under composition is formal. (Prove closed under pullback.) □

**Lemma 1.7.4.** *An affine morphism is quasi-compact.*

*Proof.* Let $f\colon X \to Y$ be an affine morphism. Then for every open immersion $\operatorname{Spec} R \to Y$, the fiber product $\operatorname{Spec} R \times_Y X$ is affine thus, by Lemma 1.6.2 also quasi-compact. □

### 1.7.2 Separated morphisms

**Definition 1.7.5.** A morphism of schemes $f\colon X \to Y$ is

- *separated* if the diagonal $\Delta_{X/Y}\colon X \to X \times_Y X$ is a closed immersion,
- *semi-separated* if the diagonal $\Delta_{X/Y}\colon X \to X \times_Y X$ is affine,
- *quasi-separated* if the diagonal $\Delta_{X/Y}\colon X \to X \times_Y X$ is quasi-compact.

**Lemma 1.7.6.** *Each of the properties separated, semi-separated and quasi-separated is closed under pullback and composition.*

*Proof.* Given a morphism $f\colon X \to Y$ with a base change $f'\colon X' \to Y'$, note that $\Delta_{X'/Y'}\colon X' \to X' \times_{Y'} X'$ is the base change of $\Delta_{X/Y}\colon X \to X \times_Y X$. Similarly, the diagonal of a composition can be written in terms of (...). Hence, these statement follow from the fact that closed immersions, affine morphisms and quasi-compact morphisms are all closed under pullback and compositions. □

**Proposition 1.7.7** (Valuative criterion of separatedness)**.** *Let* $f\colon X \to Y$ *be a morphism of schemes. The following are equivalent:*

*(i)* *$f$ is separated.*

*(ii)* *$f$ is quasi-separated and for any commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} A \arrow[r] & Y \end{tikzcd}$$

*where $A$ is a valuation ring with field of fractions $K$, there exists a morphism $\operatorname{Spec} A \to X$ completing the diagram.*

*Proof.* (Write proof)

□

### 1.7.3 Finite type morphisms

**Definition 1.7.8.** A morphism of schemes $f\colon X \to Y$ is

- *locally of finite type at $x \in |X|$* if there exists a commutative diagram

$$\begin{tikzcd} & \operatorname{Spec} k \arrow[dl] \arrow[d, "x"] \\ \operatorname{Spec} R \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} S \arrow[r] & Y \end{tikzcd}$$

  where the horizontal arrows are open immersions, such that $R$ is a finitely generated $S$-module via the ring morphism $\varphi\colon A \to B$.

- *locally of finite type* if $f$ is locally of finite type at all $x \in |X|$,

- *of finite type* if $f$ is locally of finite type and quasi-compact.

(Closed under pullback, closed under composition)

### 1.7.4 Finite presentation morphisms

**Definition 1.7.9.** A morphism of schemes $f\colon X \to Y$ is

- *locally of finite presentation at $x \in |X|$* if there exists a commutative diagram

$$\begin{tikzcd} & \operatorname{Spec} k \arrow[dl] \arrow[d, "x"] \\ \operatorname{Spec} R \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} S \arrow[r] & Y \end{tikzcd}$$

where the horizontal arrows are open immersions, such that $R$ is a finitely presented $S$-module via the ring morphism $\varphi\colon A \to B$.

- *of finite presentation* if $f$ is locally of finite presentation, quasi-compact and quasi-separated.

(Closed under pullback, closed under composition)

### 1.7.5 Proper morphisms

**Definition 1.7.10.** A morphism of schemes $f\colon X \to Y$ is *proper* if it is separated, of finite type and universally closed.

(Closed under pullback, closed under composition)

**Proposition 1.7.11** (Valuative criterion for properness)**.** *Let $f\colon X \to Y$ be a morphism of schemes. Assume $f$ is of finite type and quasi-separated. Then the following are equivalent:*

*(i) $f$ is proper.*

*(ii) For any commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} A \arrow[r] & Y \end{tikzcd}$$

*where $A$ is a valuation ring with field of fractions $K$, there exists a unique morphism $\operatorname{Spec} A \to X$ completing the diagram.*

*Proof.* (Write proof) □

### 1.7.6 Flat morphisms

**Definition 1.7.12.** A morphism of schemes $f\colon X \to Y$ is

- *flat at $x \in |X|$* if the local ring $\mathcal{O}_{X,x}$ is flat over the local ring $\mathcal{O}_{Y,|f|(x)}$,
- *flat* if $f$ is flat at all $x \in |X|$.

**Lemma 1.7.13.** *Let $f\colon X \to Y$ be a morphism of schemes. The following are equivalent.*

*(i) $f$ is flat.*

*(ii) For every commutative diagram*

$$\begin{tikzcd} \operatorname{Spec} B \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} A \arrow[r] & Y \end{tikzcd}$$

*where the horizontal arrows are open immersions, the ring $B$ is flat as an $A$-module via the ring morphism $\varphi\colon A \to B$.*

**Lemma 1.7.14.** *(i) Flat morphisms are closed under pullback.*

*(ii) Flat morphisms are closed under composition.*

*Proof.* (Write proof) *(ii)* (Find intermediate open, that could be a general lemma) *(i)* (Is the definition correct?) □

### 1.7.7 Finite morphism

**Definition 1.7.15.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is a *finite immersion* if for every commutative ring $R$ and morphism $\operatorname{Spec} R \to Y$, there is a pullback diagram

$$\begin{tikzcd} \operatorname{Spec} S \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

where $S$ is a finitely generated $R$-module via the ring morphism $\varphi\colon R \to S$.

**Lemma 1.7.16.** *Closed immersions are finite. Finite morphisms are affine.*

### 1.7.8 Integral morphism

**Definition 1.7.17.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is *integral* if for every commutative ring $R$ and morphism $\operatorname{Spec} R \to Y$, there is a pullback diagram

$$\begin{tikzcd} \operatorname{Spec} S \arrow[r] \arrow[d, "\operatorname{Spec}\varphi"'] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

where $S$ is integral over $R$ via the ring morphism $\varphi\colon R \to S$.

**Lemma 1.7.18.** *finite = integral + locally of finite type*

### 1.7.9 Surjective morphisms

**Definition 1.7.19.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is *surjective* if, for every field $k$ and element $y \in Y(k)$, there is a field extension $\ell/k$ and an element $x \in X(\ell)$ such that $f(x) = y|_\ell$.

**Lemma 1.7.20.** *Let $f\colon X \to Y$ be a morphism in $\mathbf{Set}^{\mathbf{CRing}}$. Then the following are equivalent:*

*(i) $f$ is surjective,*

*(ii) the induced map $|f|\colon |X| \to |Y|$ on topological spaces is surjective.*

*Proof.* This follows directly from the definition of the topological spaces $|X|$ and $|Y|$. □

## 1.8 Smooth, unramified and étale morphisms

**Definition 1.8.1.** A morphism $f\colon X \to Y$ of schemes is *formally smooth* (resp. *formally unramified*, *formally étale*) if for every commutative ring $R$, ideal $I \subseteq R$ with $I^2 = 0$, and commutative diagram

$$\begin{tikzcd} \operatorname{Spec} R/I \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ \operatorname{Spec} R \arrow[r] & Y \end{tikzcd}$$

there exists (resp. exists at most one, exists a unique) morphism $\operatorname{Spec} R \to X$ completing the diagram.

**Definition 1.8.2.** A morphism $f\colon X \to Y$ of schemes is

- *smooth* if it is formally smooth and locally of finite presentation,
- *unramified* if it is formally unramified and locally of finite type,
- *étale* if it is formally étale and locally of finite presentation.

**Lemma 1.8.3.** *Each of the properties (formally) smooth, (formally) unramified and (formally) étale is closed under pullback and composition.*

*Proof.* All of these are formal consequences of the definition. □

**Example 1.8.4.** When is $R \to R[x]/(f)$ (formally) smooth, étale or unramified?

**Example 1.8.5.** (Given an example of a formally étale morphism which is not étale)

### 1.8.1 Projective morphisms

(Define the different notions of projective morphisms, and relate them)

## 1.9 Projective space

**Definition 1.9.1.** Let $r, n \geq 0$ be integers. The $(r,n)$-*grassmanian* $\operatorname{Gr}(r,n)\colon \mathbf{CRing} \to \mathbf{Set}$ is the presheaf on $\mathbf{CRing}^{\mathrm{op}}$ given by

$$\operatorname{Gr}(r,n)(R) = \{M \subseteq R^n \mid M \text{ is a direct summand of rank } r\}.$$

**Proposition 1.9.2.** *For every $r, n \geq 0$, the presheaf $\operatorname{Gr}(r,n)$ is a scheme.*

*Proof.* (Write proof) □

**Definition 1.9.3.** For any $n \geq 0$, *projective $n$-space*, denoted $\mathbb{P}^n$, is the scheme $\operatorname{Gr}(1, n+1)$.

(More generally, construct a functor $\operatorname{Proj}\colon (\mathbf{CRing}^{\mathrm{op}})^{\mathbb{Z}} \to \mathbf{Sch}$)

## 1.10 Sheaves and modules over schemes

Let $X$ be a scheme. By a *sheaf on* $X$ we refer to a sheaf on the category $\mathbf{Aff}_X$ with respect to the Zariski topology.

**Definition 1.10.1.** Let $X$ be a scheme. A *module* over $X$ is a sheaf $\mathcal{F}$ of abelian groups on $X$, together with, for every $x\colon \operatorname{Spec} R \to X$, the structure of an $R$-module on $\mathcal{F}(x)$, such that, for every $f\colon (R', x') \to (R, x)$, the induced morphism $\mathcal{F}(f)\colon \mathcal{F}(x) \to \mathcal{F}(x')$ is $R$-linear, viewing $\mathcal{F}(x')$ as an $R$-module via the map $R \to R'$.

A morphism of modules over $X$ is a morphism of sheaves $\mathcal{F} \to \mathcal{F}'$ which respects the module structures. Denote the category of modules over $X$ by $\mathbf{Mod}_X$.

**Definition 1.10.2.** Let $X$ be a scheme. A module $\mathcal{F}$ over $X$ is *quasi-coherent* if, for all $f\colon (\operatorname{Spec} R', x') \to (\operatorname{Spec} R, x)$ the induced map $\mathcal{F}(R, x) \otimes_R R' \to \mathcal{F}(R', x')$ is an isomorphism. Denote the full subcategory of quasi-coherent modules over $X$ by $\mathbf{QCoh}_X$.

**Example 1.10.3.** When $X = \operatorname{Spec} R$, we have an equivalence of categories $\mathbf{QCoh}_X \simeq \mathbf{Mod}_R$. Any $R$-module $M$ induces (functorially) a quasi-coherent module over $X$, via $\mathcal{F}(x) = M \otimes_R R'$ for all $x\colon \operatorname{Spec} R' \to \operatorname{Spec} R$. Conversely, given a module $\mathcal{F}$ over $X$, the identity morphism $\mathrm{id}_X$ retrieves the module $M$.

**Definition 1.10.4.** Let $X$ be a scheme. The *structure sheaf* of $X$, denoted $\mathcal{O}_X$, is the quasi-coherent module over $X$ given by $\mathcal{O}_X(R, x) = R$.

**Definition 1.10.5.** Let $X$ be a scheme. A module $\mathcal{F}$ over $X$ is *coherent* if it is quasi-coherent and if $\mathcal{F}(x)$ is a finitely presented $R$-module for every $x\colon \operatorname{Spec} R \to X$. Denote the full subcategory of coherent modules over $X$ by $\mathbf{Coh}_X$.

**Proposition 1.10.6.** *Let $X$ be a scheme. The categories $\mathbf{Mod}_X$, $\mathbf{QCoh}_X$ and $\mathbf{Coh}_X$ are abelian.*

*Proof.* (Write proof) □

**Definition 1.10.7.** Let $X$ be a scheme. A module $\mathcal{F}$ over $X$ is *locally free* if $\mathcal{F}(x)$ is a projective $R$-module for all $x\colon \operatorname{Spec} R \to X$. A module $\mathcal{F}$ over $X$ is *free of rank $n$* if it is isomorphic to $\bigoplus_{i=1}^n \mathcal{O}_X$.

**Definition 1.10.8.** Let $f\colon X \to Y$ be a morphism of schemes. Given a module $\mathcal{F}$ over $Y$, define the *pullback* of $\mathcal{F}$ along $f$ as the module $f^*\mathcal{F}$ over $X$ given by

$$(f^*\mathcal{F})(R, x) = \mathcal{F}(R, f \circ x).$$

This construction yields additive functors $f^*\colon \mathbf{Mod}_Y \to \mathbf{Mod}_X$ and $f^*\colon \mathbf{QCoh}_Y \to \mathbf{QCoh}_X$.

(Might want to extend any module over $X$ to a sheaf on the whole of $\mathbf{Sch}_X$)

**Definition 1.10.9.** Let $f\colon X \to Y$ be a morphism of schemes. Given a module $\mathcal{F}$ over $X$, define the *direct image* of $\mathcal{F}$ along $f$ as the module $f_*\mathcal{F}$ over $Y$ given by

$$(f_*\mathcal{F})(R, y) = \lim_{\operatorname{Spec} S \xrightarrow{z} X \times_Y \operatorname{Spec} R} \mathcal{F}(S, \pi_X \circ z).$$

This construction yields additive functors $f_*\colon \mathbf{Mod}_X \to \mathbf{Mod}_Y$ and $f_*\colon \mathbf{QCoh}_X \to \mathbf{QCoh}_Y$.

**Proposition 1.10.10.** *For any morphism $f\colon X \to Y$ of schemes, $f^*$ is left adjoint to $f_*$.*

*Proof.* (Write proof) □

**Definition 1.10.11.** Let $X$ be a scheme. An *ideal* over $X$ is a module $\mathcal{I}$ over $X$ such that $\mathcal{I}(R, x) \subseteq R$ is an ideal for every $\operatorname{Spec} R \xrightarrow{x} X$.

**Proposition 1.10.12.** *Let $X$ be a scheme. There is an equivalence of categories between the category of quasi-coherent ideals over $X$ and the category of closed immersions to $X$.*

*Proof.* Any closed immersion $i\colon Z \to X$ determines a quasi-coherent ideal $\mathcal{I}$ over $X$ as follows. For any $\operatorname{Spec} R \xrightarrow{x} X$, the pullback of $i$ along $x$ is given by $\operatorname{Spec} S \to \operatorname{Spec} R$ for some surjective ring morphism $\varphi\colon R \to S$, and we put $\mathcal{I}(R, x) = \ker \varphi$. For any ring morphism $(R', x') \to (R, x)$, the pullback of $i$ along $x'$ is given by $\varphi'\colon R' \to S'$ with $S' = S \otimes_R R'$, so there is a natural morphism $\mathcal{I}(R, x) \to \mathcal{I}(R', x')$ and the induced morphism $\mathcal{I}(R, x) \otimes_R R' \to \mathcal{I}(R', x')$ is indeed an isomorphism. Hence, $\mathcal{I}$ is a quasi-coherent ideal over $X$. (Check the sheaf condition) Conversely, a quasi-coherent ideal $\mathcal{I}$ over $X$ determines the presheaf $Z\colon \mathbf{CRing} \to \mathbf{Set}$ given by

$$Z(R) = \{x \in X(R) \mid \mathcal{I}(R, x) = 0\}$$

and a morphism $i\colon Z \to X$ given by the inclusion. Now, clearly $i$ is a closed immersion, and hence $Z$ is a scheme by Corollary 1.4.8. One verifies that everything is functorial and that the two constructions are inverse to each other. □

## 1.11 Stalks

(In this section, we could ignore the prime ideals, by localizing with respect to the multiplicatively closed set $S = f^{-1}(k^\times)$)

Let $X$ be a scheme. Note that, by Proposition 1.4.4, for every field $k$ and element $x \in X(k)$, the morphism $\operatorname{Spec} k \xrightarrow{x} X$ factors as $\operatorname{Spec} k \xrightarrow{x'} \operatorname{Spec} R \overset{i}{\hookrightarrow} X$ for some open immersion $i$.

**Definition 1.11.1.** Let $X$ be a scheme and $\mathcal{F}$ a module over $X$. Given a field $k$ and element $x \in X(k)$, factor $x$ as $\operatorname{Spec} k \xrightarrow{x'} \operatorname{Spec} R \overset{i}{\hookrightarrow} X$ for some open immersion $i$. The *stalk* of $\mathcal{F}$ at $x$, denoted $\mathcal{F}_x$, is defined as the localization $\mathcal{F}(R, i) \otimes_R R_{\mathfrak{p}}$, where $\mathfrak{p}$ is the kernel of $x'\colon R \to k$. In the particular, the stalk $\mathcal{O}_{X,x}$ is equal to $R_{\mathfrak{p}}$. The stalk $\mathcal{F}_x$ naturally admits the structure of an $\mathcal{O}_{X,x}$-module.

**Lemma 1.11.2.** *The above definition does not depend (up to isomorphism) on the choice of factorization through* $\operatorname{Spec} R$.

*Proof.* Suppose $x$ factors both as $\operatorname{Spec} k \xrightarrow{x'} \operatorname{Spec} R \overset{i}{\hookrightarrow} X$ and as $\operatorname{Spec} k \xrightarrow{x''} \operatorname{Spec} R' \overset{i'}{\hookrightarrow} X$. Taking the fiber product of $\operatorname{Spec} R$ and $\operatorname{Spec} R'$ over $X$, and applying Proposition 1.4.4 again, both $x'$ and $x''$ factor through some open immersions $\operatorname{Spec} S \overset{j}{\hookrightarrow} \operatorname{Spec} R$ and $\operatorname{Spec} S \overset{j'}{\hookrightarrow} R'$, respectively. Writing $R \xrightarrow{\psi} S \xrightarrow{\varphi} k$ for the corresponding morphisms on rings, it suffices to show that $\psi$ induces an isomorphism $R_{\mathfrak{p}} \cong S_{\mathfrak{q}}$, where $\mathfrak{q} = \ker \varphi$ and $\mathfrak{p} = \ker(\psi \circ \varphi)$. (Prove this) $\square$

## 1.12 Differentials

**Definition 1.12.1.** Let $\varphi\colon A \to B$ be a morphism of commutative rings, and let $M$ be a $B$-module. An *$A$-derivation* of $B$ is an additive map $d\colon B \to M$ such that $d(\varphi(a)) = 0$ for all $a \in A$ and $d(bb') = bd(b') + b'd(b)$ for all $b, b' \in B$.

**Definition 1.12.2.** Let $\varphi\colon A \to B$ be a morphism of commutative rings. A *module of relative differentials* of $B$ over $A$ is the $B$-module $\Omega_{B/A}$, together with an $A$-derivation $d\colon B \to \Omega_{B/A}$, which has the following universal property: for any $B$-module $M$ and $A$-derivation $d'\colon B \to M$, there exists a unique $B$-module morphism $f\colon \Omega_{B/A} \to M$ such that $d' = f \circ d$. Note that, if a module of relative differentials of $B$ over $A$ exists, it is unique up to isomorphism.

**Proposition 1.12.3.** *Let $\varphi\colon A \to B$ be a morphism of commutative rings. Define $f\colon B \otimes_A B \to B$ by $f(b \otimes b') = bb'$ and let $I = \ker f$. Considering $B \otimes_A B$ as a $B$-module by multiplication on the left, $I/I^2$ inherits the structure of a $B$-module. Define $d\colon B \to I/I^2$ by $d(b) = 1 \otimes b - b \otimes 1$. Then $I/I^2$ with $d$ is a module of relative differentials for $B/A$.*

*Proof.* Indeed, $d$ is additive, $d(\varphi(a)) = 0$ for all $a \in A$, and $d(bb') - bd(b') - b'd(b) = (b \otimes 1 - 1 \otimes b)(b' \otimes 1 - 1 \otimes b') \equiv 0 \mod I^2$, so $d$ is an $A$-derivation. (Finish proof) $\square$

**Example 1.12.4.** When $B = A[x_1, \ldots, x_n]/(f_1, \ldots, f_m)$, then $\Omega_{B/A}$ is generated by $dx_1, \ldots, dx_m$ modulo the relations $df_i = 0$. (Proof this)

(Recall differentials for commutative rings)

(Define a sheaf of differentials $\Omega_{X/Y}$ on $X$ for any morphism $f\colon X \to Y$ of schemes.)

(Show that the stalk of the sheaf of differentials is the dual of the tangent space)

## 1.13 Sheaf cohomology

(Note: sheaves of abelian groups are not defined over $X$ yet, $\mathbf{Sh}_X$.)

**Definition 1.13.1.** Let $X$ be a scheme. For any sheaf $\mathcal{F}$ over $X$, let

$$\Gamma(X, \mathcal{F}) = \lim_{(R,x) \in \mathbf{Aff}_X} \mathcal{F}(x),$$

considered as an abelian group. This construction is functorial in $\mathcal{F}$, and the resulting functor

$$\Gamma(X, -) \colon \mathbf{Sh}_X \to \mathbf{Ab}$$

is called the *global sections functor*.

(After defining the global sections functor, and showing enough injectives exist, we obtain right derived functors. It might be nice to construct all six-functors.)

# Chapter 2

# Algebraic spaces

## 2.1 The fpqc topology

The letters 'fpqc' stand for *fidèlement plate et quasi-compacte*.

**Definition 2.1.1.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is an *fpqc cover* if for every commutative ring $R$ and element $y \in Y(R)$, there exists a faithfully flat ring morphism $\varphi\colon R \to S$ and an element $x \in X(S)$ such that $f(S)(x) = Y(\varphi)(y)$. (Check if this is correct)

**Proposition 2.1.2.** *(i) Any isomorphism $f\colon X \to Y$ is an fpqc cover.*

*(ii) The class of fpqc covers is closed under composition: for all fpqc covers $f\colon X \to Y$ and $g\colon Y \to Z$, also $g \circ f$ is an fpqc cover.*

*(iii) If $g \circ f$ is an fpqc cover, then so is $g$.*

*(iv) The class of fpqc covers is closed under pullbacks: for any fpqc cover $f\colon X \to Y$ and morphism $g\colon Z \to Y$, also $X \times_Y Z \to Z$ is an fpqc cover.*

*(v) If $f\colon X \to Y$ is an fpqc cover and $g\colon Z \to Y$ is a morphism such that $Z \times_Y \operatorname{Spec} R \to \operatorname{Spec} R$ is an fpqc cover for all commutative rings $R$ and $x \in X(R)$, then $g$ is also an fpqc cover.*

*Proof.* *(i)* is obvious. *(ii)* follows from the fact that the composition of faithfully flat morphisms is again faithfully flat. *(iii)* is straightforward. *(iv)* is proven formally. *(v)* (Might split up (v) into two statements: (1) if $f\colon X \to Y$ is such that $X \times_Y \operatorname{Spec} R \to \operatorname{Spec} R$ is fpqc for all $x \in Y(R)$, then $f$ is fpqc itself, and (2) if the base change of a morphism along an fpqc morphism is fpqc, then the morphism is fpqc itself.) □

**Definition 2.1.3.** The *fpqc topology* on $\mathbf{CRing}^{\mathrm{op}}$ is defined to be the Grothendieck topology whose covering sieves for a commutative ring $R$ is the collection of subfunctors $U \subseteq \operatorname{Spec} R$ such that the inclusion is an fqpc cover. By Proposition 2.1.2, this indeed defines a Grothendieck topology.

The goal of the remainder of this section is to prove the following theorem.

**Theorem 2.1.4.** *Any scheme $X\colon \mathbf{CRing} \to \mathbf{Set}$ is a sheaf in the fpqc topology.*

**Lemma 2.1.5.** *Let $R \to S$ be a faithfully flat ring morphism. Then the sequence $0 \to R \to S \to S \otimes_R S$, where the latter map is $s \mapsto s \otimes 1 - 1 \otimes s$, is exact (as a sequence of $R$-modules).*

*Proof.* Since faithfully flat morphisms are injective, the sequence is exact at $R$. As $R \to S$ is faithfully flat, exactness at $S$ is equivalent to the exactness of

$$S \xrightarrow[\varphi]{s \mapsto s \otimes 1} S \otimes_R S \xrightarrow[\psi]{s \otimes t \mapsto s \otimes t \otimes 1 - s \otimes 1 \otimes t} S \otimes_R S \otimes_R S.$$

Clearly, $\operatorname{im} \varphi \subseteq \ker \psi$. Take an element $x = \sum_{i=1}^n s_i \otimes t_i$ of $\ker \psi$. Applying the map $S \otimes_R S \otimes_R S \to S \otimes_R S$ given by $r \otimes s \otimes t \mapsto rs \otimes t$ to $\psi(x) = 0$ yields $x = (\sum_{i=1}^n s_i t_i) \otimes 1$, which shows $x \in \operatorname{im} \varphi$. □

*Proof of Theorem 2.1.4.* (Reduce somehow to Lemma 2.1.5) □

## 2.2 The fppf topology

The letters 'fppf' stand for *fidèlement plate de présentation finie*.

**Definition 2.2.1.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is an *fppf cover* if for every commutative ring $R$ and element $y \in Y(R)$, there exists a faithfully flat ring morphism $\varphi\colon R \to S$ of finite presentation and an element $x \in X(S)$ such that $f(S)(x) = Y(\varphi)(y)$.(Check if this is correct)

**Lemma 2.2.2.** *Any fppf cover is an fpqc cover.*

*Proof.* (Write proof) □

## 2.3 The étale topology

**Definition 2.3.1.** A morphism $f\colon X \to Y$ in $\mathbf{Set}^{\mathbf{CRing}}$ is an *étale cover* if for every commutative ring $R$ and element $y \in Y(R)$, there exists an étale $R$-algebra $\varphi\colon R \to S$ and an element $x \in X(S)$ such that $\operatorname{Spec} S \to \operatorname{Spec} R$ is surjective and $f(S)(x) = Y(\varphi)(y)$. (Check if this is correct)

**Proposition 2.3.2.** *(i)* (Properties étale covers)

**Definition 2.3.3.** The *étale topology* on $\mathbf{CRing}^{\mathrm{op}}$ is defined to be the Grothendieck topology whose covering sieves for a commutative ring $R$ is the collection of subfunctors $U \subseteq \operatorname{Spec} R$ such that the inclusion is an étale cover. By Proposition 2.3.2, this indeed defines a Grothendieck topology.

**Remark 2.3.4.** Every Zariski cover is an étale cover. Hence, the étale topology is finer than the Zariski topology.

## 2.4 Algebraic spaces

**Definition 2.4.1.** An *algebraic space* is an étale sheaf $X\colon \mathbf{CRing} \to \mathbf{Set}$ such that

- the diagonal $X \to X \times X$ is representable by schemes, that is, for every scheme $S$ and morphism $S \to X$, the fiber product $X \times_{X \times X} S$ is a scheme,
- there exists a scheme $U$ and a morphism $U \to X$ which is étale and surjective.

**Lemma 2.4.2.** *Every scheme is an algebraic space.*

*Proof.* Every scheme $X$ is a sheaf in the fpqc topology by Theorem 2.1.4, and hence an étale sheaf. The diagonal $X \to X \times X$ is clearly representable by schemes, and the identity morphism $X \to X$ is étale and surjective. □

**Example 2.4.3.** (Example of an algebraic space which is not a scheme.)

# Chapter 3

# Algebraic stacks

(Write this chapter somewhere in the future)